BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 151 · Paper 2: Validation, Testing and Debugging

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does selection do?

Q2. What is NOT true?
A. true
B. 0.5
C. false
D. An empty string

Q3. “A Boolean variable represents any unrestricted paragraph of text.” Correct this. Explain why.

Q4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

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I Do · Read new knowledge

Key vocabulary

Validation — Test data against defined rules and loop back after rejection if further input is required.

Testing — Include normal, boundary, invalid and erroneous inputs.

Logic errors — A syntactically valid program may still use incorrect conditions or updates.

Maintainability — Simple named variables, comments and subprograms support future debugging.

New learning

Validation: Test data against defined rules and loop back after rejection if further input is required.

Testing: Include normal, boundary, invalid and erroneous inputs.

Logic errors: A syntactically valid program may still use incorrect conditions or updates.

Maintainability: Simple named variables, comments and subprograms support future debugging.

Worked example

A test plan is needed for a quantity 1–20 inclusive. Produce four categories.

1. Normal quantity=10.

2. Boundary quantities=1 and 20.

3. Invalid numeric quantities 0 and 21.

4. Erroneous input 'many' should be handled safely.

We Do · Practise with guidance

Guided 1

Fix a quantity check using quantity>=1 OR quantity<=20.

Worked method: Accept 1 and 20, reject 0 and 21.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Fix the range check for an image width of zero pixels.

```
width = 0
if width >= 1 OR width <= 1920 then
    print("Valid")
else
    print("Invalid")
endif
```

iterate

Question 1

Give one invalid and one erroneous input for age 5–16. [2 marks]

Question 2

Explain the difference between validation and authentication. [2 marks]

Question 3

Describe how a trace table helps locate a logic error. [3 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Boundary testing includes the smallest and largest valid values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It chooses a branch using a condition. Selection can be nested.

Q2. C — false

Q3. True or false.

Q4. (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules.

I Do / We Do pages · Guided practice

Q1. Accept 1 and 20, reject 0 and 21.

You Do pages · Code First

```
width = 0
if width >= 1 AND width <= 1920 then
    print("Valid")
else
    print("Invalid")
endif
```

Expected result: Invalid

You Do pages · Independent questions

Q1. Invalid 20 (integer out of range); erroneous 'cat' (non-numeric text).

Q2. Validation checks input constraints; authentication checks claimed identity.

Q3. It records successive states and shows the first variable/branch differing from expected behaviour.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: